

Ornithopter Flapping Wing Mechanism

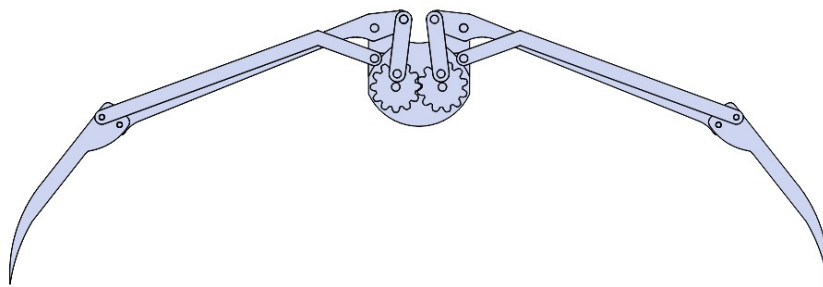


Fig. 1 – 2D CAD Design of Ornithopter Mechanism

Subject	Engineering Design & Fabrication
Project Title	Ornithopter Mechanism
Design Tool	SolidWorks Premium 2024 SP1.0
Fabrication	3D Printing & Wooden Frame
Date	May 2026

1. Introduction

An ornithopter is a flying mechanism that generates lift and thrust by flapping its wings, mimicking the natural flight of birds and insects. This project was undertaken to design, model, and fabricate a working ornithopter prototype that demonstrates the fundamental principles of bio-inspired mechanical engineering.

The project integrates computer-aided design using SolidWorks with physical fabrication using 3D printing and wooden materials. The core engineering challenge involves converting continuous rotary motion — from a motor or manual input — into synchronized oscillating (flapping) motion of two wings, connected through a gear-and-linkage mechanism.

2. Objectives

- To study and understand the working principle of an ornithopter flapping mechanism.
- To design a symmetric, functional flapping wing assembly using SolidWorks.
- To understand motion transmission through gear systems and four-bar linkages.
- To fabricate a working prototype using 3D printed components and wooden structural elements.
- To validate the CAD model against the physical prototype through testing and observation.

3. Materials Used

Component	Material / Type	Purpose
Base Platform	Wooden board	Structural support for the entire assembly
Wing Arms (Ulna)	3D Printed PLA	Left and right wing structural members
Wing Tips	3D Printed PLA	Curved claw-shaped wingtip elements
Metacarpus Bracket	3D Printed PLA	Central bracket connecting gears to wings
Gears (x2)	3D Printed PLA	Transmit and synchronize rotary motion
Radius Links	3D Printed PLA	Upper linkages connecting gear to wing
Pivot Joints	Screws / Nails	Allow rotation at articulation points
Adhesive	Glue	Bonding and stability of joints

4. CAD Design (SolidWorks)

The ornithopter mechanism was fully designed in SolidWorks Premium 2024 SP1.0 as a multi-body assembly. Each sub-component — including the base board, metacarpus bracket, radius links, ulna arms, and 3D printed gear set — was modelled individually and assembled using appropriate mates (concentric, coincident, and hinge mates) to replicate real joint behavior.

The assembly tree includes the following named components: Base Board (01), Metacarpus (02), Radius links (03), Left Ulna (04A), Right Ulna (04B), and the interlocking gear pair at the center. Motion studies were conducted in SolidWorks to verify synchronized wing oscillation before physical fabrication.

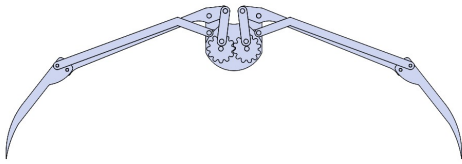


Fig. 2 – Front View (2D Layout)

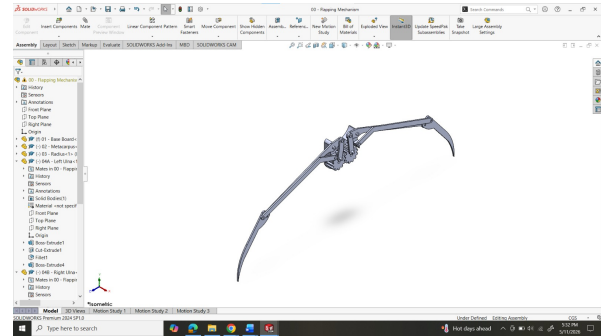


Fig. 3 – Isometric 3D Assembly View

5. Construction of the Model

The physical model was fabricated following the SolidWorks design. The construction process involved two parallel streams: 3D printing of the functional mechanical components, and manual preparation of the wooden base structure.

Step 1 – 3D Printing: All mechanical components (wings, gears, brackets, and links) were sliced and printed in PLA using an FDM 3D printer. Post-processing included light sanding of gear teeth for smooth meshing.

Step 2 – Base Preparation: A wooden block was cut and finished to serve as the mounting base. Holes were pre-drilled to align with the gear shaft and mounting screws.

Step 3 – Gear Assembly: The two interlocking spur gears were mounted on their respective shafts at the center of the base. Alignment was carefully verified to ensure smooth meshing and equal torque transmission.

Step 4 – Linkage Assembly: Radius links were attached to the gear cranks at offset pivot points. The opposite ends of the links were pinned to the upper ends of the ulna arm assemblies.

Step 5 – Wing Attachment: Left and right ulna arms were pivot-mounted to the metacarpus bracket. Wing-tip elements were attached at the far ends, completing the wing structure.

Step 6 – Testing & Adjustment: Manual rotation of the gear shaft was used to verify synchronized flapping. Minor adjustments were made to pivot tightness to achieve smooth, symmetric motion.



Fig. 4 – Physical Prototype (3D Printed Wings on Wooden Base)

6. Working Principle

The ornithopter mechanism operates on the principle of converting continuous rotary motion into oscillatory (reciprocating) angular motion. The central spur gear pair acts as the primary motion

source. When one gear is driven (manually or by a motor), the meshed gear counter-rotates at an equal speed.

Crank pins offset from the gear centers drive the radius links in a circular path. These links push and pull the upper pivot points of the ulna arms, causing each wing to swing upward and downward in alternation. Since the cranks on the two gears are phased symmetrically, both wings flap in perfect synchronization — rising and falling together as seen in birds during soaring flight.

Stage	Component	Motion Type
1	Gear Input (manual/motor)	Continuous Rotation
2	Spur Gear Pair	Counter-Rotation (1:1 ratio)
3	Crank Pin on Gear	Circular (eccentric) Motion
4	Radius Link	Linear Reciprocation
5	Ulna Wing Arm	Oscillating Angular Motion
6	Wingtip	Flapping Up & Down

7. Mechanism Analysis

The mechanism combines three classical systems:

Gear Mechanism: Two meshing spur gears transmit rotary motion and synchronize the phase of both wings. The 1:1 gear ratio ensures equal flapping frequency on both sides.

Crank-Rocker (Four-Bar) Linkage: The crank (gear + offset pin), connecting rod (radius link), and rocker (ulna arm) form a four-bar linkage. This is the core mechanism converting rotation to oscillation.

Oscillating Wing Mechanism: The ulna arms pivot about fixed points on the metacarpus bracket, producing the characteristic bird-like flapping arc at the wingtips.

8. Advantages

- Bio-inspired design closely replicates natural bird flight kinematics.
- Lightweight 3D printed PLA components reduce overall structural weight.
- Simple gear-linkage system requires minimal maintenance.
- Modular assembly allows easy replacement of individual components.
- Economical fabrication using accessible materials (wood + 3D printing).
- Clear motion conversion makes it an excellent educational demonstration model.

9. Observations & Testing

The fabricated prototype was tested through repeated manual actuation of the gear shaft. The following observations were recorded:

- Wings moved smoothly and symmetrically with each rotation of the gear shaft.
- Proper gear tooth alignment was critical — misalignment caused binding and uneven motion.
- 3D printed PLA parts provided sufficient rigidity while remaining lightweight.
- Wooden base offered stable support with no flex or vibration during operation.
- The claw-shaped wingtips traced a consistent arc, confirming accurate linkage geometry.
- The physical prototype closely matched the motion behavior predicted in the SolidWorks Motion Study.

10. Applications

Educational Models: Demonstrates gear systems, linkage mechanisms, and motion conversion in mechanical engineering courses.

Robotic Bird / UAV: Scaled-up versions can serve as flapping-wing unmanned aerial vehicles (ornithopter UAVs) for surveillance or research.

Bio-inspired Robotics: Forms the basis for soft-robotics research in animal-like locomotion systems.

Toy & Display Models: Motorized versions make interactive display exhibits for science fairs and museums.

Flight Research: Provides a physical test bed for studying aerodynamic efficiency of flapping vs. fixed-wing flight.

11. Conclusion

The Ornithopter Flapping Wing Mechanism project successfully achieved all its design and fabrication objectives. Starting from first-principles understanding of bird flight, the team designed a complete mechanical assembly in SolidWorks, verified its motion through digital simulation, and fabricated a working physical prototype using 3D printing and wooden materials.

The final prototype clearly demonstrates the conversion of rotary motion into synchronized wing flapping through a gear-driven four-bar linkage system. The project deepened practical understanding of mechanical linkages, gear geometry, CAD assembly techniques, and bio-inspired engineering design — skills directly applicable to robotics, aerospace, and product design fields.

— End of Report —